CISS350: Data Structures and Advanced Algorithms
Quiz q10802

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You are given this class for the row of a hashtable:

```
const int Available = 0;
const int NotAvailable = 1;
const int Deleted = 2;

class HashtableRow
{
public:
    HashtableRow(int flag, int stu_id, double height)
        : flag_(flag), stu_id_(std_id), height_(height)
    {}
    int flag_;
    int stu_id_;
    double height_;
};

int hash(int x)
{
    return 2 * x + 3;
}
```

Complete the following function to insert a row into the given hashtable. This uses non-separate chaining. The index of the row where the insert occurs is returned. If the table is full, -1 is returned. n is the size of the hashtable.
Answer:

```
int hashtable_insert(HashtableRow hashtable[], int n, int key, double value)
{
  int index = -1;
  int h = hash(key) % n;
  int start = h;

  while(true)
  {
    if(hashtable[h].flag == 2) // Deleted
      index = h
    else if(hashtable[h].flag == 0) // Available
    {
      if(index == -1)
        index = h;

      hashtable[h].stu_id_ = key;
```

```
      hashtable[h].height_ = value;
      hashtable[h].flag = 1;

      return index;
    }
    else if(hashtable[h].flag == 1) // Not Available
    {
      if(hashtable[h].stu_id_ == key)
        return -1;
    }

    h = (h + 1) % n;

    if(h == start)
      return -1;
  }
}
```

Instructions

In the file `thispreamble.tex` look for

```
\renewcommand\AUTHOR{}
```

and enter your email address:

```
\renewcommand\AUTHOR{jdoe5@cougars.ccis.edu}
```

(This is not really necessary since alex will change that for you when you execute `make`.) In your bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

Enter your answers in `main.tex`. In the bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

For each question, you'll see boxes for you to fill. For small boxes, if you see

```
1 + 1 = \answerbox{}.
```

you do this:

```
1 + 1 = \answerbox{2}.
```

`answerbox` will also appear in "true/false" and "multiple-choice" questions.

For longer answers that need typewriter font, if you see

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
\end{answercode}
```

you do this:

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
int x;
\end{answercode}
```

`answercode` will appear in questions asking for code, algorithm, and program output. In this case, indentation and spacing is significant. For program output, I do look at spaces and newlines.

For long answers (not in typewriter font) if you see

```
What is the color of the sky?
\begin{answerlong}
\end{answerlong}
```

you can write

```
What is the color of the sky?
\begin{answerlong}
The color of the sky is blue.
\end{answerlong}
```

A question that begins with "T or F or M" requires you to identify whether it is true or false, or meaningless. "Meaningless" means something's wrong with the question and it is not well-defined. Something like "$1 + 2 = 4$" is either true or false (of course it's false). Something like "$1+_2 = 4$?" does not make sense.

When writing results of computations, make sure it's simplified. For instance write 2 instead of $1 + 1$.

Higher level classes.

For students beyond 245: You can put LaTeX commands in `answerlong`.

More examples of meaningless statements: Questions such as "Is $42 = 1+_2$ true or false?" or "Is $42 = \{2\}^{\{3\}}$ true or false?" does not make sense. "Is $P(42) = \{42\}$ true or false?" is meaningless because $P(X)$ is only defined if X is a set. For "Is $1 + 2 + 3$ true or false?", "$1 + 2 + 3$" is well–defined but as a "numerical expression", not as a "proposition", i.e., it cannot be true or false. Therefore "Is $1 + 2 + 3$ true or false?" is also not a well-defined question.

More examples of simplification: When you write down sets, if the answer is $\{1\}$, do not write $\{1, 1\}$. And when the values can be ordered, write the elements of the set in ascending order. When writing polynomials, begin with the highest degree term.

When writing a counterexample, always write the simplest.